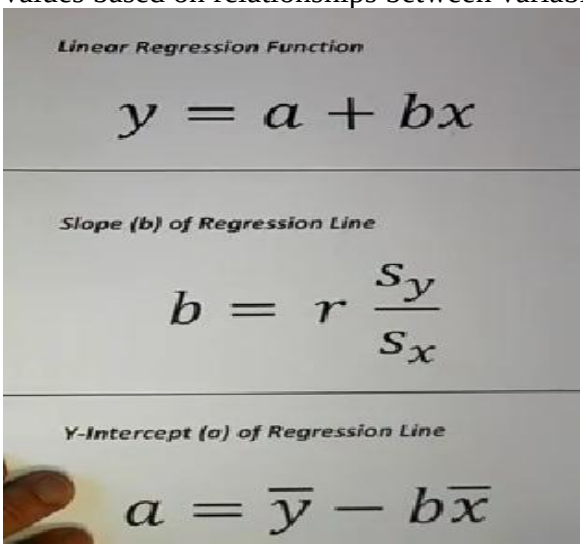
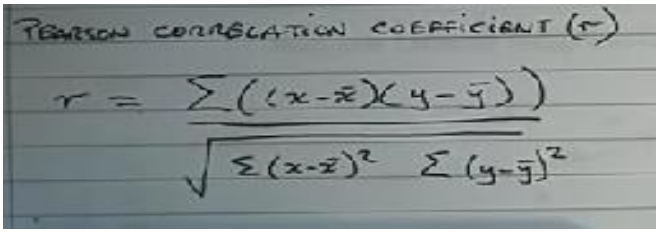
ISelec 3 Reviewer

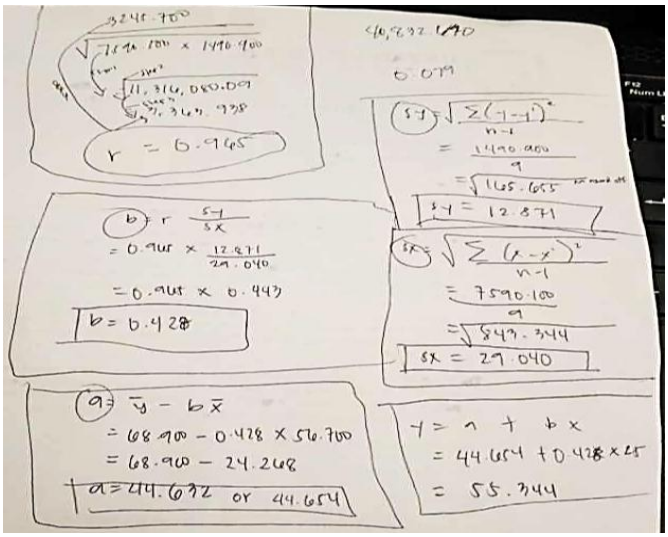
Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict continuous values based on relationships between variables.



No.	Assignment Score	Exam Score	x-x'	y-y'	(x-x')*(y-y')	(x-x')^2	(y-y')^2
1	12	50	-44.700	-18.900	844.830	1998.090	357.210
2	23	55	-33.700	-13.900	468.430	1135.690	193.210
3	31	58	-25.700	-10.900	280.130	660.490	118.810
4	45	60	-11.700	-8.900	104.130	136.890	79.210
5	53	72	-3.700	3.100	-11.470	13.690	9.610
6	62	69	5.300	0.100	0.530	28.090	0.010
7	71	78	14.300	9.100	130.130	204.490	82.810
8	84	74	27.300	5.100	139.230	745.290	26.010
9	89	85	32.300	16.100	520.030	1043.290	259.210
10	97	88	40.300	19.100	769.730	1624.090	364.810
	56.700	68.900			3245.700	7590.100	1490.900
r	0.965	b	0.428				
Sy	12.871	a	44.654				
Sx	29.040	y	when x=25	25	55.344		
			when x=65	65	72.449		
			when x=80	80	78.864		





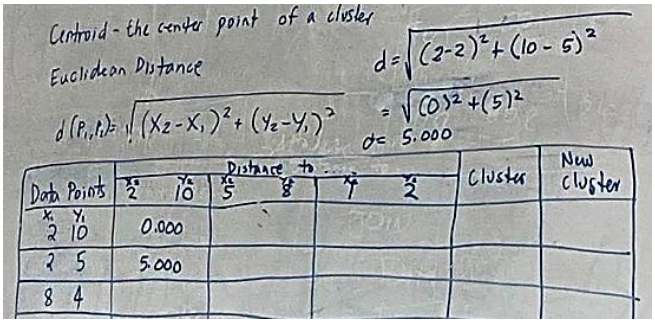
K-MEANS – an algorithm that group data path based on K (clusters)

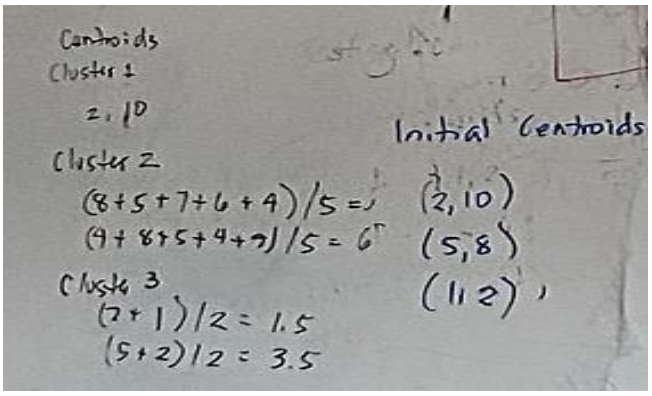
CLUSTERING- process of grouping similar objects into clusters

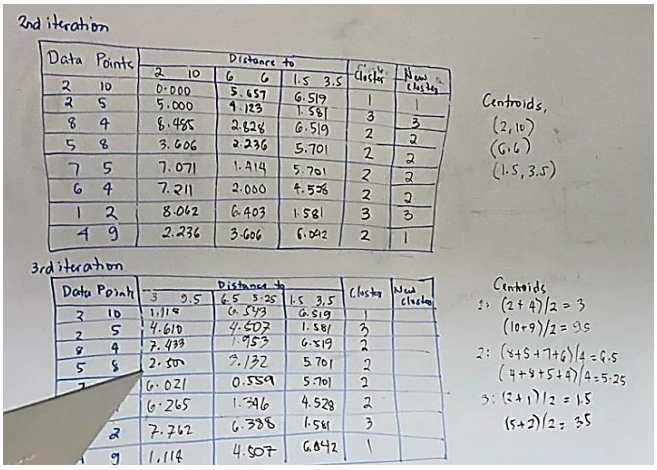
CENTROID- the center point of a cluster

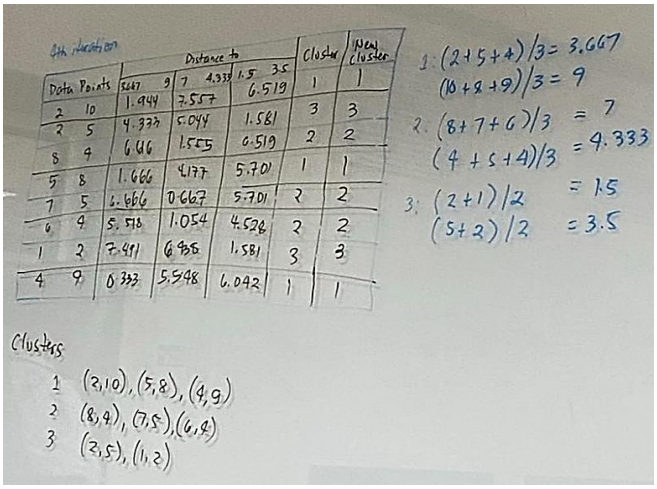
EUCLIDEAN DISTANCE

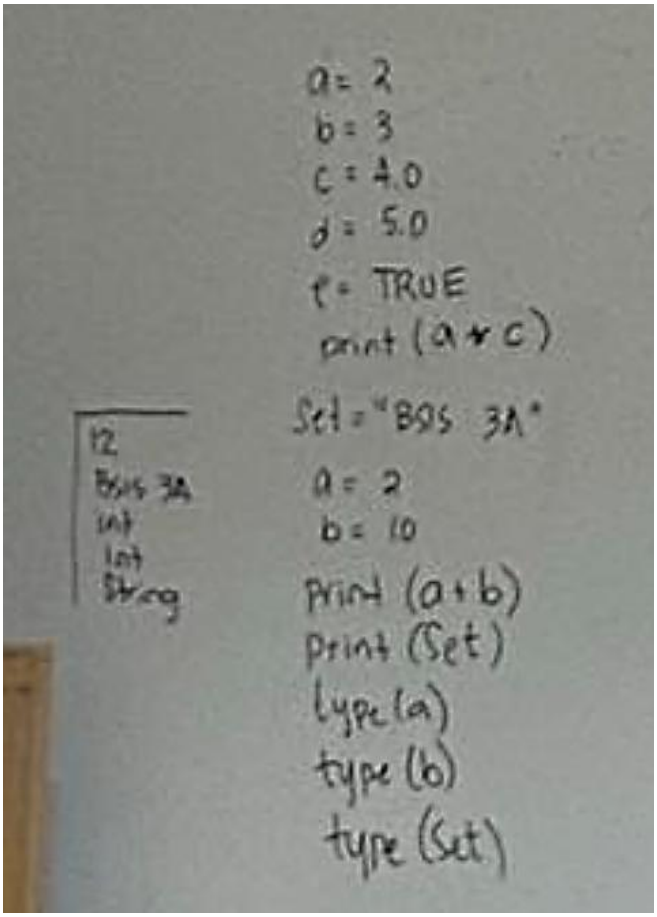
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$











PHYTON PROGRAMMING

– is a high-level programming language developed by Guido Van Rossum

-officially released last 1991

DATA TYPE IN PYTHON

Int, float, complex, str, bool, list, tuple, dict, set, None Type

ARITHMETIC OPERATIONS

+, -, *, /, % (modulo division), ** (exponentiation)

BACKLASH CODE/ ESCAPE SEQUENCE

\n new line

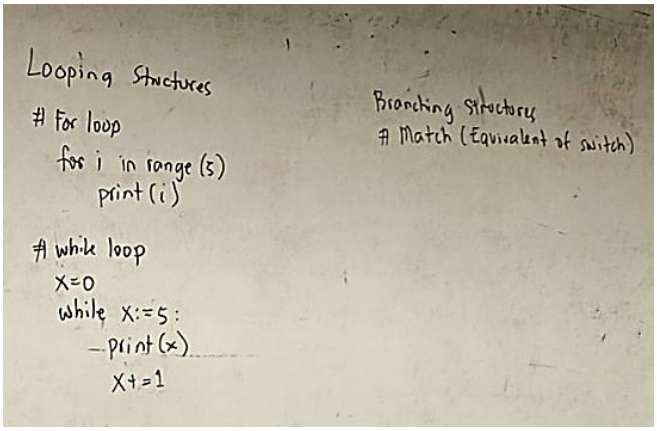
\t tab space

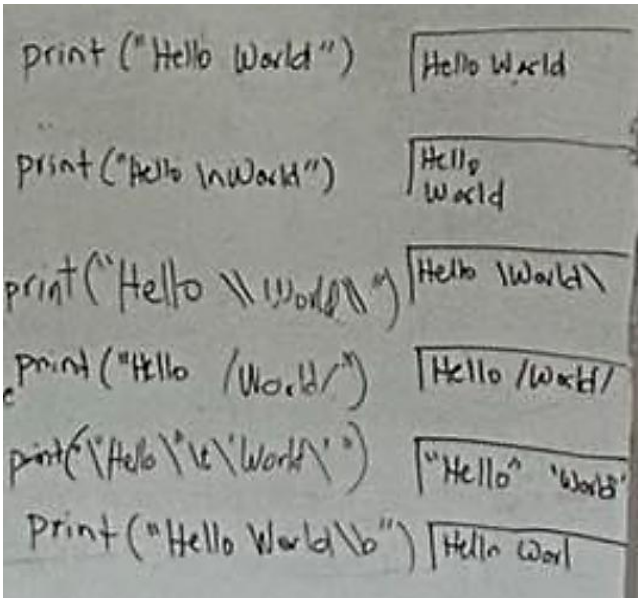
\\

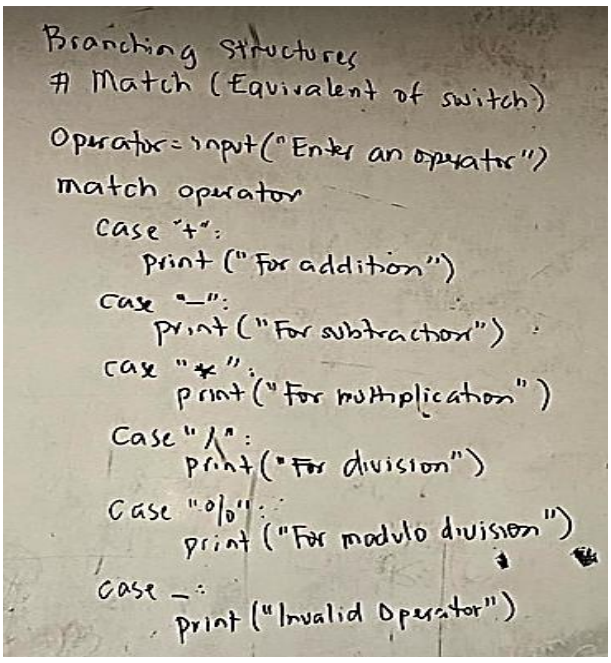
\\

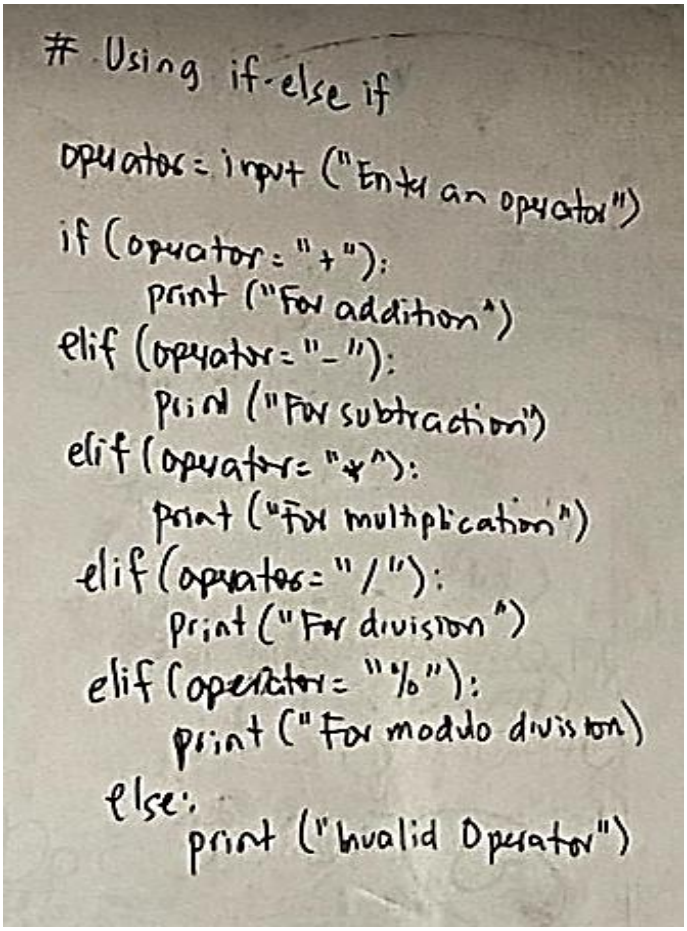
\”

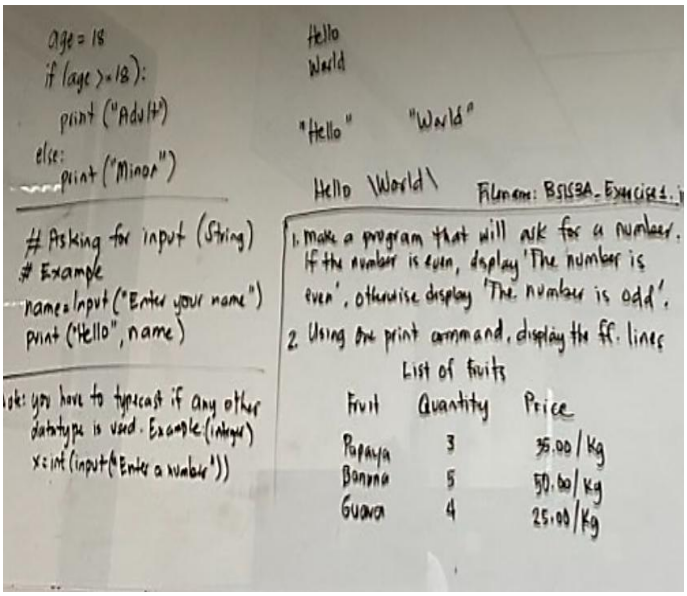
\b backspace

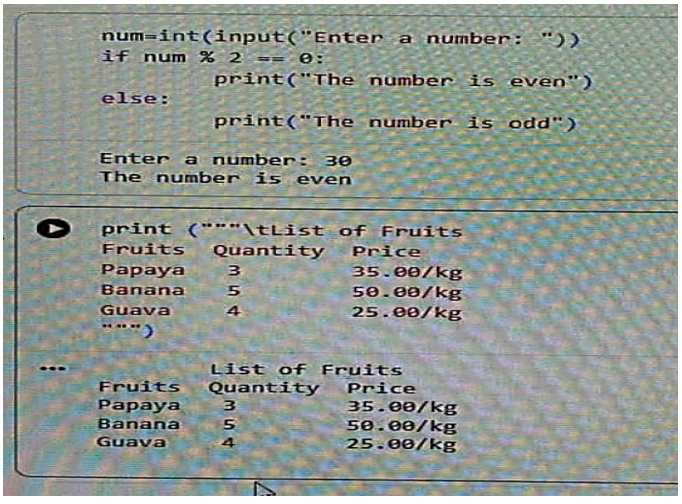












1. What will be the output of the following Python code snippet?

```
x = [1, 2, 3]
y = x
y.append(4)
print(x)
```

ANS. [1, 2, 3, 4]

2. Which of the following is the correct way to define a function in Python?

ANS. Def my_func():

3. What is the primary difference between a list and a tuple in Python?

ANS. List are mutable, while tuples are immutable.

4. What will be the result of executing `bool("")` in Python?

ANS. False

5. How do you start a multi-line comment in Python?

ANS. Python does not have a formal multi-line comment syntax; you use multiple # symbols or unassigned multi-line strings

6. What does the `len()` function return when passed a dictionary?

ANS. The total number of keys in the dictionary

7. What is the output of `print(2 3)` in Python? (exponentation)

ANS. 8

8. Which keyword is used to handle exceptions or errors in Python?

ANS. Except

9. What will `"hello".upper()` return?

ANS. HELLO

10. What is the purpose of the `break` statement in a loop?

ANS. To terminate the loop completely and resume execution at the next statement outside the loop.

CONFUSION MATRIX- A Confusion Matrix is used to evaluate classification models.

It compares:

- Actual values
- Predicted values

Structure of Confusion Matrix

	Predicted Positive	**Predicted Negative**
Actual Positive	TP	FN
Actual Negative	FP	TN

Meaning

- **TP (True Positive)**
  Correctly predicted positive
- **TN (True Negative)**
  Correctly predicted negative
- **FP (False Positive)**
  Incorrectly predicted positive
- **FN (False Negative)**
  Incorrectly predicted negative

Example

	Predicted Yes	Predicted No
Actual Yes	40	10
Actual No	5	45

Here:

- TP = 40
- FN = 10
- FP = 5
- TN = 45

PERFORMANCE METRICS (ACCURACY, PRECISIOM, RECALL, F1-SCORE)

These metrics measure how good a classification model performs.

Accuracy

Formula

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Meaning

Percentage of correct predictions.

Example

If 90 out of 100 predictions are correct:

- Accuracy = 90%

Precision

Formula

$$Precision = \frac{TP}{TP+FP}$$

Meaning

Measures how many predicted positives are actually correct.

High Precision

Few false positives.

Recall

Formula

$$Recall = \frac{TP}{TP+FN}$$

Meaning

Measures how many actual positives are correctly identified.

High Recall

Few false negatives.

F1-Score

Formula

$$F1 = 2\left(\frac{Precision \times Recall}{Precision + Recall}\right)$$

Meaning

Balance between Precision and Recall.

Use Case

Useful when dataset is imbalanced.

Quick Comparison Table

Metric	Focus
Accuracy	Overall correctness
Precision	Correct positive predictions
Recall	Finding all positives
F1-score	Balance of precision & recall